CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2012

EC 301 Electromagnetic Theory

Date: 26.11.2012, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. ***Make suitable assumptions*** and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. ***Be precise in giving answers as per marks weight.***

SECTION – I

Q - 1 (a) Provide general expression of a vector in Cartesian Coordinate system in three dimensions. Also provide equations of magnitude of vector and associated unit vector in Cartesian Coordinate System. **[03]**

(b) Given points A($x = 2$, $y = 3$, $z = -1$) and B($\rho = 4$, $\phi = -50^0$, $z = 2$), find the distance from: (i) A to origin; (ii) B to the origin; (iii) A to B. **[04]**

Q - 2 (a) Using neat sketch, define Position vector. In addition, illustrate and explain divergence, gradient and curl with example. **[04]**

(b) Calculate **E** at M (3,-4,2) in free space caused by: (i) Charge $Q_1 = 2\mu C$ at $P_1(0,0,0)$ (ii) a charge $Q_2 = 3\mu C$ at $P_2(-1,2,3)$ (iii) a charge $Q_1 = 2\mu C$ at P_1 (0,0,0) and a charge $Q_2 = 3\ \mu C$ at P_2 (-1,2,3). **[05]**

(c) Find **D** in Cartesian coordinates at P(6,8,-10) caused by (i) a point charge of 30 mC at the origin (ii) A uniform line charge $\rho_L = 40\ \mu C/m$ on the z axis; (iii) A uniform surface charge density $\rho_s = 57.2\ \mu C/m^2$ on the plane $x = 9$. **[05]**

OR

Q - 2 (a) Give expression for the cross product in spherical coordinates between two vectors and between unit vectors. Compare divergence and gradient using example. **[04]**

(b) Four infinite uniform sheets of charge are located as follows: 20 pC/m^2 at $y = 7$, -8 pC/m^2 at $y = 3$, 6 pc/m^2 at $y = -1$ and -18 pC/m^2 at $y = -4$. Find **E** at the point: (i) $P_A(2,6,-4)$; (ii) $P_B(0,0,0)$; (iii) $P_C(-1,-1,1.5)$. **[05]**

(c) Surface charge densities of 200, -50 and ρ_{sx} $\mu C/m^2$ are located at $r = 3$, 5, and 7 cm respectively. Find **D** at r =: (i) 2 cm (ii) 4 cm. Also, Find ρ_{sx} if **D** =0 at $r = 7.32$ cm. **[05]**

Q - 3 (a) Explain divergence theorem. Derive equation of divergence of a vector. Mention divergence D of any vector in terms of rectangular, cylindrical and spherical forms. [04]

(b) Illustrating neat diagram, explain Strokes' theorem. In addition, explain magnetic flux density. [04]

(c) With neat diagram, explain boundary conditions for perfect dielectric materials [06]

OR

Q - 3 (a) Explain electric field intensity. Derive equation of electric field intensity for charge Q in three dimension system. In addition, mention the equation of Electric field intensity for ***n*** combined point charges. [04]

(b) With neat figure, describe ampere circuital law. [04]

(c) With neat diagram separating two magnetic materials, explain boundary conditions for perfect magnetic material. [06]

SECTION – II

Q - 4 (a) Provide two bifurcations between Diamagnetic, Paramagnetic and Ferromagnetic Materials. [03]

(b) Mention Lorentz force equation. With neat figure, explain force on a differential current element. [04]

Q - 5 (a) The magnetic flux density in a magnetic material with $\chi_m = 6$ is given in a certain region as $\mathbf{B} = 0.005\, y^2 \mathbf{a}_x$ T. At y = 0.4 m, find the magnitude of (i) $\mathbf{J}$ (ii) $\mathbf{J}_b$ (iii) $\mathbf{J}_T$ [02]

(b) Show that the solution of Laplace's equation is unique. [04]

(c) Describe Displacement current in brief. Derive updated equation of Ampere Circuital law which includes the displacement current. [04]

OR

Q - 5 (a) If $\mathbf{B} = 0.05\, x\mathbf{a}_y$ T in a material for which $\chi_m = 2.5$ Find (i) μ_R (ii) μ (iii) $\mathbf{M}$ (iv) $\mathbf{J}$ [02]

(b) Derive point form of continuity equation. [04]

(c) Derive any one of the Maxwell's equation in point form. [04]

Q - 6 Attempt Any Three **[18]**

(i) Describe Standing Wave Ratio with neat diagram and necessary equations

(ii) Starting with vector Laplacian of $\mathbf{E}_s$, develop the intrinsic impedance of free space as 377 Ω

(iii) Derive the equation of Poynting Vector.

(iv) Derive: Inductance $L = \dfrac{\mu_0 N^2 S}{2\pi\rho_0}$

(v) Providing equation, briefly mention physical interpretation of depth of penetration. In addition, explain wave polarization in brief. Also, using Electric field in xy plane in diagram, explain linear polarization, left circular polarization (l.c.p), right circular polarization (r.c.p.) and elliptical polarization in brief.

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